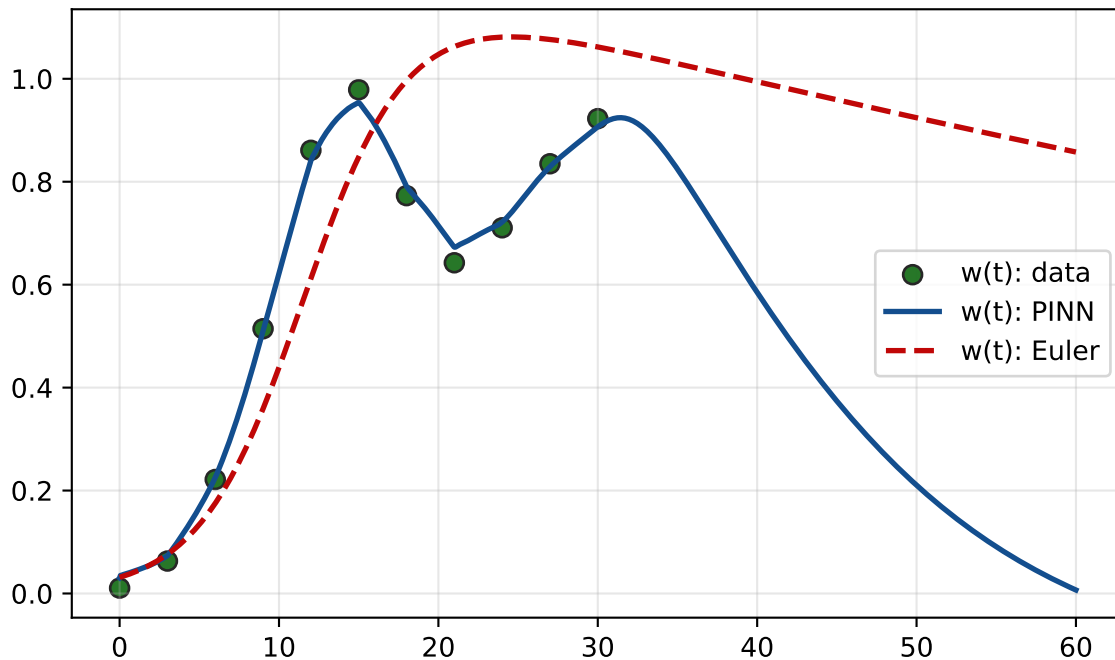
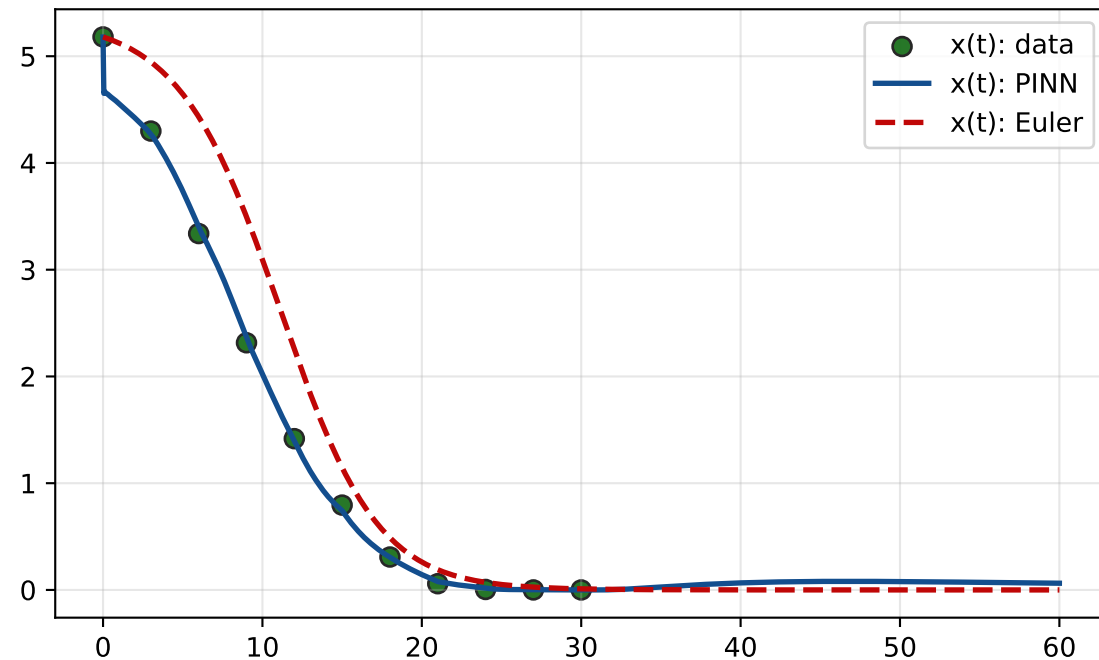


# E8 | PINN vs Euler

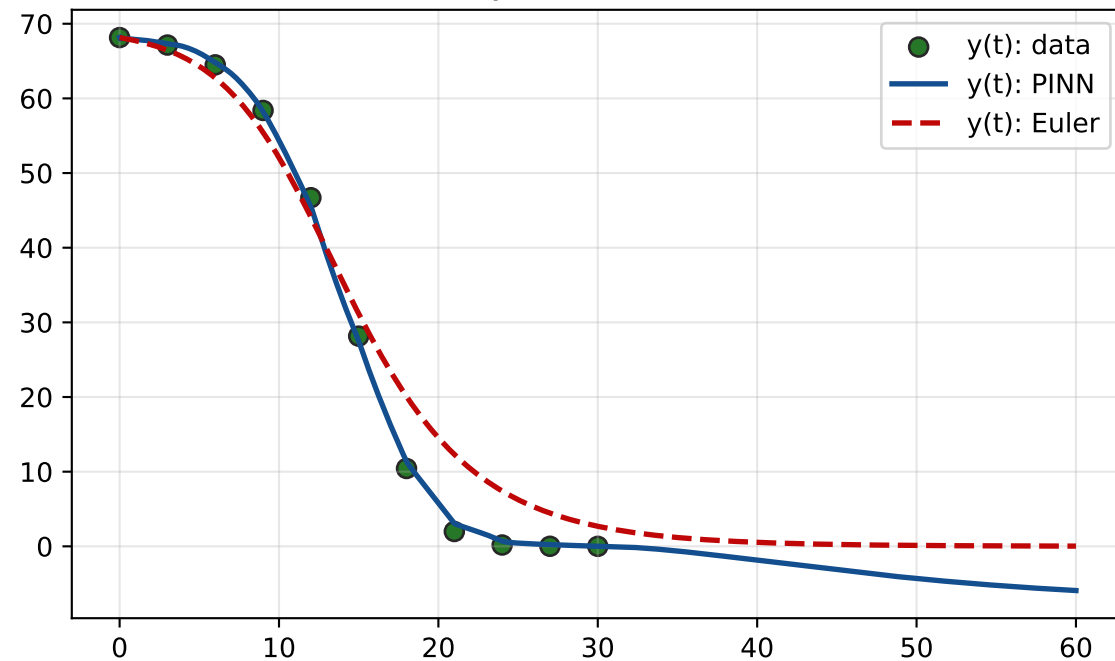
w(t): Biomass



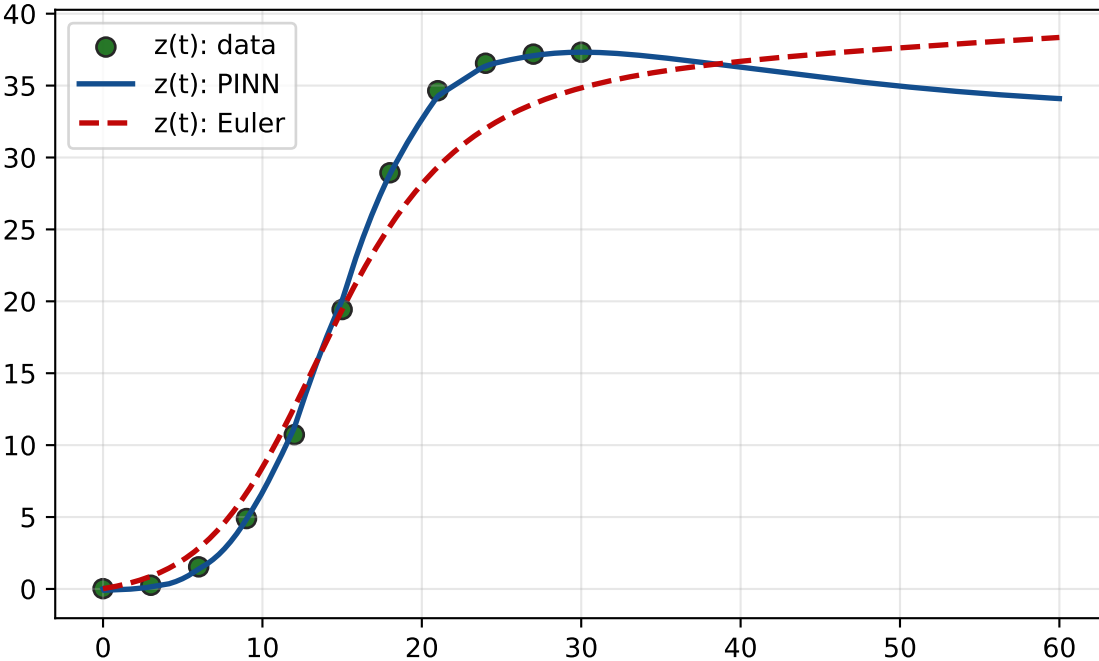
x(t): Glucose



y(t): Fructose



z(t): Ethanol



E8

R2=0.9788, R2\_adj=0.9746, RSS=4.005e+02, AIC=1.143e+02

cond(JTJ)=1.06e+07, used\_pinv=False

| param | estimate     | SE           | MOE_95       | CI_low        | CI_high      | p_value      |
|-------|--------------|--------------|--------------|---------------|--------------|--------------|
| p1    | 4.119343e-03 | 1.607045e-03 | 3.256183e-03 | 8.631598e-04  | 7.375526e-03 | 1.456725e-02 |
| p2    | 9.997214e-02 | 2.927922e-01 | 5.932534e-01 | -4.932813e-01 | 6.932256e-01 | 7.347010e-01 |
| p3    | 1.918081e-04 | 3.870231e-03 | 7.841832e-03 | -7.650024e-03 | 8.033640e-03 | 9.607397e-01 |
| p4    | 3.052111e-01 | 3.694699e-01 | 7.486172e-01 | -4.434061e-01 | 1.053828e+00 | 4.140548e-01 |
| p6    | 1.570486e-01 | 1.111862e-01 | 2.252846e-01 | -6.823604e-02 | 3.823332e-01 | 1.661643e-01 |
| p8    | 7.712652e-02 | 5.553348e-02 | 1.125215e-01 | -3.539500e-02 | 1.896480e-01 | 1.731886e-01 |
| p9    | 1.647973e-05 | 1.380351e-02 | 2.796857e-02 | -2.795209e-02 | 2.798505e-02 | 9.990538e-01 |

| IC             | estimate     |
|----------------|--------------|
| w0 (estimated) | 3.132480e-02 |
| x0 (fixed)     | 5.180769e+00 |
| y0 (fixed)     | 6.814856e+01 |
| z0 (fixed)     | 1.921567e-02 |
| W0 (fixed)     | 0.000000e+00 |